



Hotel Performance Analysis

Operational Insights Dashboard: Visualizing Performance Trends for Hotel Bookings



About the Project

This project presents an end-to-end data workflow designed to transform raw, inconsistent hotel booking records into actionable business insights. It covers hotel performance across three major U.S. cities: San Francisco, Los Angeles, and New York.

Part A: Data Integrity Audit & Optimization

*This phase focuses on cleaning and preparing over **22,000** multi-source records. It involves standardizing inconsistent formats, handling missing and invalid values, removing duplicates, and validating computed fields. The result is a structured and reliable dataset ready for analysis.*

Part B: Operational Insights Dashboard

Building on the cleaned dataset, this phase delivers an interactive dashboard that visualizes key performance trends. Users can dynamically explore hotel booking data by city and year through slicers, enabling a clear and focused view of operational patterns.

This project demonstrates how raw data can be systematically transformed into meaningful insights through structured cleaning and effective visualization.

Part A | [Full Process](#) & [Output](#)

Part B | [Output](#) & [Full Report](#)

Hotel Performance Analysis

booking_id	guest_name	email	listing_id	listing_city	checkin_date	checkin_mon	checkin_year	checkout_date	night	price_per_night	total_price	payment_status	payment_status
KG-013035	Unknown	nhoffman@scott.com	1083	Los Angeles	24/05/2023	May	2023	12/12/2023	202	\$245.61	\$49,613.22	Pending	At Risk
KG-003115	Carolyn Shah	achan@olson-nelson.com	1079	Los Angeles	19/08/2024	Aug	2024	08/11/2024	81	\$51.96	\$4,208.76	NA	At Risk
KG-008732	Jermaine Huang	thomasadam@yahoo.com	1036	San Francisco	20/03/2024	Mar	2024	03/03/2025	348	\$383.63	\$133,503.24	Paid	Safe
KG-007591	Amy Rios	reginald31@robertson-smith.com	1025	Los Angeles	29/02/2024	Feb	2024	16/03/2025	381	\$268.36	\$102,245.16	Pending	At Risk
KG-000221	Unknown	NA	1089	Los Angeles	26/01/2025	Jan	2025	22/02/2025	27	\$134.71	\$3,637.17	Pending	At Risk
KG-017295	Unknown	nichole43@gmail.com	1038	Los Angeles	03/02/2025	Feb	2025	16/02/2025	13	\$383.22	\$4,981.86	Unpaid	At Risk
KG-005857	Unknown	brianbrown@lynn-taylor.com	1066	San Francisco	25/06/2024	Jun	2024	19/10/2024	116	\$283.24	\$32,855.84	NA	At Risk
KG-012608	Amanda Martinez	susanpham@livingston.com	1035	Los Angeles	21/03/2025	Mar	2025	28/03/2025	7	\$189.12	\$1,323.84	Pending	At Risk
KG-003463	Unknown	eharrison@mullins.com	1038	New York	10/08/2023	Aug	2023	21/11/2023	103	\$392.45	\$40,422.35	Unpaid	At Risk
KG-006478	Shelby Mason	grahamamanda@sparks.com	1089	San Francisco	19/10/2024	Oct	2024	14/04/2025	177	\$195.82	\$34,660.14	Pending	At Risk
KG-003350	Hunter Weber	erinsmith@smith-kim.net	1081	San Francisco	30/04/2024	Apr	2024	12/06/2024	43	\$94.88	\$4,079.84	Paid	Safe
KG-015982	Elijah Osborne	madisongray@jacobson.org	1021	San Francisco	16/04/2025	Apr	2025	15/05/2025	29	\$464.95	\$13,483.55	Paid	Safe
KG-019275	James Lester	hannah00@yahoo.com	1044	Los Angeles	20/10/2023	Oct	2023	23/10/2023	3	\$466.35	\$1,399.05	Paid	Safe
KG-018089	Unknown	NA	1074	San Francisco	26/08/2023	Aug	2023	08/03/2025	560	\$189.67	\$106,215.20	Paid	Safe
KG-018753	Robert Reid	dcastro@hotmail.com	1058	New York	23/12/2023	Dec	2023	23/02/2025	428	\$268.87	\$115,076.36	Paid	Safe
KG-011503	Unknown	jonesshawn@yahoo.com	1035	New York	04/02/2024	Feb	2024	12/02/2024	8	\$342.45	\$2,739.60	Pending	At Risk
KG-004058	Unknown	michael88@gibson-burch.net	1083	New York	07/09/2024	Sep	2024	01/12/2024	85	\$418.30	\$35,555.50	Paid	Safe
KG-004874	Lisa Richard	NA	1006	Los Angeles	13/11/2023	Nov	2023	02/01/2024	50	\$286.15	\$14,307.50	Paid	Safe
KG-019936	Jared Jordan	ellenlloyd@perez.biz	1044	San Francisco	18/07/2024	Jul	2024	02/04/2025	258	\$466.93	\$120,467.94	Pending	At Risk
KG-014055	Tammie Hall	ellisstephanie@gutierrez.net	1092	New York	19/08/2023	Aug	2023	12/02/2023	NA	\$273.47		Paid	Safe
KG-016852	Erica Smith	rodriguezjames@hayes.com	1095	New York	19/11/2023	Nov	2023	25/03/2025	492	\$278.03	\$136,790.76	NA	At Risk
KG-018250	Linda Perry	taylorwheeler@yahoo.com	1081	Los Angeles	31/08/2023	Aug	2023	03/01/2024	125	\$67.05	\$8,381.25	Paid	Safe
KG-015813	Margaret McClain	NA	1099	Los Angeles	06/11/2023	Nov	2023	05/11/2024	365	\$461.04	\$168,279.60	Pending	At Risk
KG-016414	John Werner Dvm	brayjoseph@gmail.com	1040	San Francisco	07/08/2024	Aug	2024	12/04/2025	248	\$329.64	\$81,750.72	Paid	Safe
KG-019453	Robert Cunningham	becky06@hotmail.com	1085	Los Angeles	24/09/2024	Sep	2024	21/01/2025	119	\$457.42	\$54,432.98	Paid	Safe
KG-001489	Chad Moore	douglas18@harmon.com	1067	Los Angeles	08/11/2024	Nov	2024	08/11/2024	0	\$95.22	\$0.00	NA	At Risk
KG-011967	John Torres	nmoss@gmail.com	1090	New York	28/02/2024	Feb	2024	01/05/2025	428	\$480.46	\$205,636.88	Pending	At Risk
KG-009561	Michael Davis	johnsonsabrina@yahoo.com	1072	San Francisco	03/03/2025	Mar	2025	03/03/2025	0	\$237.31	\$0.00	NA	At Risk
KG-001398	Bryce Gutierrez	vickiwood@gmail.com	1093	New York	31/01/2024	Jan	2024	27/09/2024	240	\$419.89	\$100,773.60	NA	At Risk
KG-013829	William Fisher	christopherbrown@yahoo.com	1083	San Francisco	08/03/2025	Mar	2025	17/05/2025	70	\$326.59	\$22,861.30	Paid	Safe
KG-012431	Jennifer Ortiz	palmerrebecca@sanford-mckenz	1090	New York	14/08/2024	Aug	2024	13/04/2025	242	\$149.06	\$36,072.52	Pending	At Risk
KG-016160	Megan Grant	sharon41@yahoo.com	1067	Los Angeles	18/04/2024	Apr	2024	27/04/2025	374	\$87.84	\$32,852.16	Paid	Safe
KG-008688	Unknown	debbieowens@willis.com	1001	San Francisco	17/03/2024	Mar	2024	17/03/2024	0	\$351.59	\$0.00	NA	At Risk
KG-001056	Sheila Miller	stacy94@robertson.info	1059	New York	24/03/2024	Mar	2024	30/04/2024	37	\$443.81	\$16,420.97	Pending	At Risk
KG-014736	Sean Ross	james32@harris.biz	1082	New York	23/10/2024	Oct	2024	07/11/2024	15	\$0.00	\$0.00	Free	Safe
KG-000945	William Howard	rachel86@brown.org	1081	San Francisco	10/01/2024	Jan	2024	31/12/2024	356	\$445.75	\$158,687.00	Pending	At Risk
KG-011832	Jasmine Savage	megannpalmer@ioseph.com	1075	Los Angeles	21/12/2024	Dec	2024	31/03/2025	100	\$430.22	\$43,022.00	NA	At Risk



Hotel
Performance
Analysis



Hotel Performance Analysis Calculations

Total Revenue			
\$	721,362,734		
Sum of total_price			
	Los Angeles	New York	Grand Total
2023			
Apr	\$6,556,972.72	\$10,495,585.72	\$17,052,558.44
May	\$27,000,459.18	\$33,138,935.63	\$60,139,394.81
Jun	\$21,154,260.97	\$29,269,578.32	\$50,423,839.29
Jul	\$19,898,096.06	\$29,405,408.84	\$49,303,504.90
Aug	\$23,849,239.16	\$27,372,314.06	\$51,221,553.22
Sep	\$16,456,212.67	\$28,135,401.01	\$44,591,613.68
Oct	\$19,000,970.71	\$26,039,957.14	\$45,040,927.85
Nov	\$19,256,469.61	\$24,847,238.47	\$44,103,708.08
Dec	\$15,713,897.26	\$21,371,340.20	\$37,085,237.46
2024			
Jan	\$15,617,350.75	\$22,321,528.74	\$37,938,879.49
Feb	\$15,403,171.22	\$17,324,984.34	\$32,728,155.56
Mar	\$14,556,785.29	\$21,331,896.88	\$35,888,682.17
Apr	\$13,515,406.95	\$17,886,574.83	\$31,401,981.78
May	\$12,366,685.99	\$16,468,863.80	\$28,835,549.79
Jun	\$9,333,321.11	\$15,581,072.19	\$24,914,393.30
Jul	\$11,475,144.14	\$12,596,417.04	\$24,071,561.18
Aug	\$9,915,126.06	\$13,259,789.09	\$23,174,915.15
Sep	\$7,178,010.22	\$10,815,606.15	\$17,993,616.37
Oct	\$6,896,146.80	\$9,269,605.13	\$16,165,751.93
Nov	\$5,280,898.82	\$8,164,556.85	\$13,445,455.67
Dec	\$5,078,581.06	\$6,961,999.10	\$12,040,580.16
2025			
Jan	\$4,028,801.10	\$5,702,856.41	\$9,731,657.51
Feb	\$2,849,590.15	\$3,773,566.19	\$6,623,156.34
Mar	\$2,463,308.19	\$2,808,783.95	\$5,272,092.14
Apr	\$938,558.78	\$1,235,409.35	\$2,173,968.13
May	\$0.00	\$0.00	\$0.00
Grand Total	\$305,783,464.97	\$415,579,269.43	\$721,362,734.40

Booking Volume			
14,001			
Count of booking_id	Column Labels		
Row Labels	Los Angeles	New York	Grand Total
2023	2034	2775	4809
Apr	73	116	189
May	276	340	616
Jun	243	293	536
Jul	237	325	562
Aug	268	338	606
Sep	200	345	545
Oct	253	360	613
Nov	245	331	576
Dec	239	327	566
2024	2991	4009	7000
Jan	248	336	584
Feb	230	310	540
Mar	244	355	599
Apr	254	342	596
May	268	342	610
Jun	233	334	567
Jul	279	326	605
Aug	272	363	635
Sep	234	346	580
Oct	269	323	592
Nov	232	317	549
Dec	228	315	543
2025	949	1243	2192
Jan	256	337	593
Feb	230	299	529
Mar	274	337	611
Apr	181	261	442
May	8	9	17
Grand Total	5974	8027	14001

Average Price		
\$	272.25	196
Row Labels		
Sum of nights	Average Price	
2023		
Apr	66179	
May	229865	
Jun	188604	
Jul	183988	
Aug	188552	
Sep	163696	
Oct	171265	
Nov	161404	
Dec	143173	
2024		
Jan	140898	
Feb	126174	
Mar	135758	
Apr	113912	
May	109060	
Jun	93805	
Jul	89989	
Aug	89517	
Sep	70251	
Oct	62549	
Nov	48869	
Dec	43101	
2025		
Jan	36217	
Feb	25222	
Mar	19929	
Apr	8444	
May	0	
Grand Total	2710421	

Hotel Performance Analysis Dashboard

Los Angeles, New York | 2023 - 2025

Search here...

City

Los Angeles

New York

San Francisco

Years

2023

2024

2025

Note: For more reliable results, this analysis focuses on yearly and city-wide performance rather than monthly fluctuations.

Total Revenue
\$721,362,734

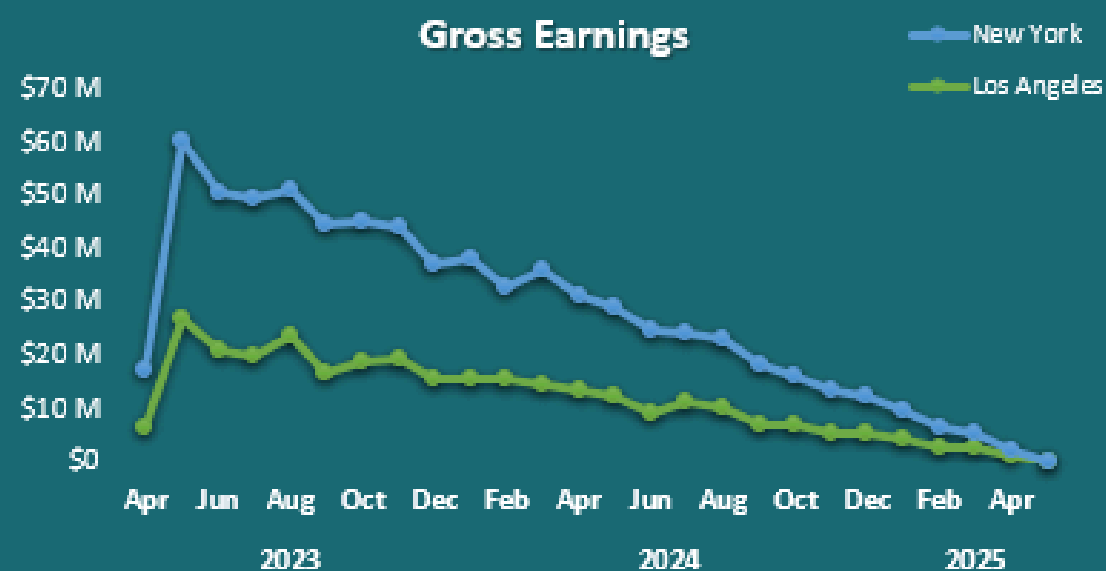
Total Bookings
14,001

Review Score
4.52

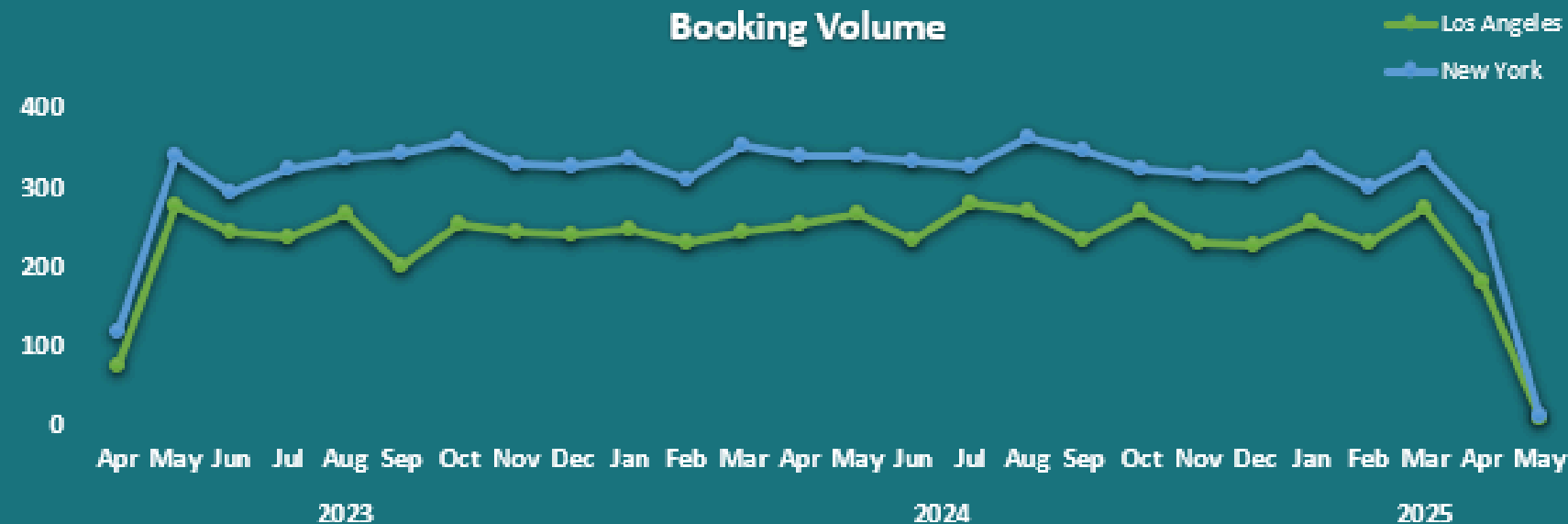
Average Price
\$272.25

Average Stay
196

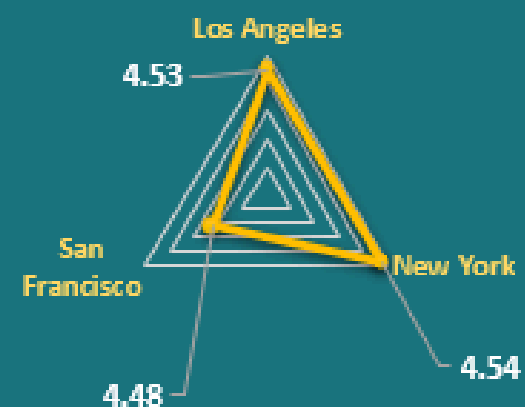
Gross Earnings



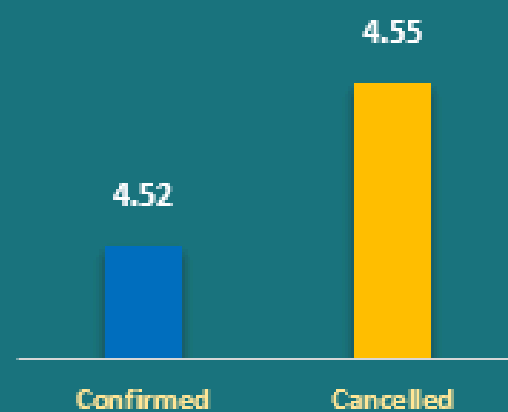
Booking Volume



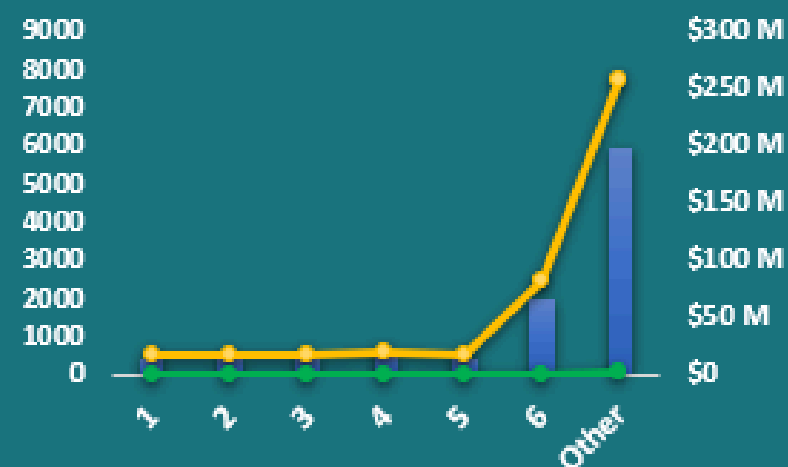
Guest Satisfaction



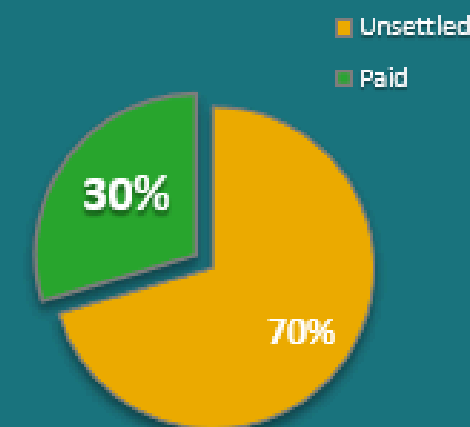
Review Correlation



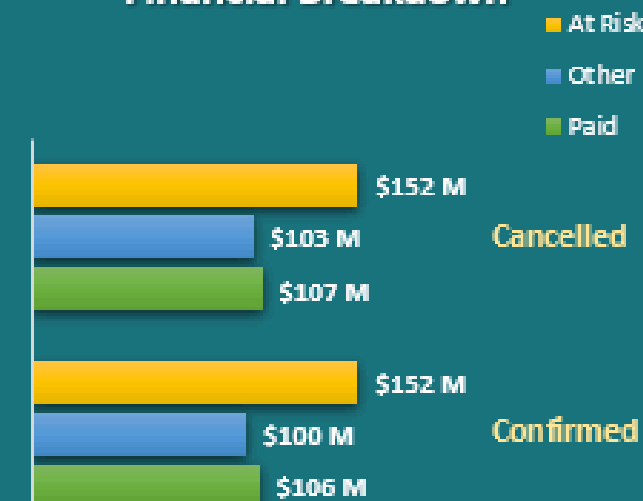
Quality vs. Cash



Settled Transaction



Financial Breakdown



Hotel Performance Analysis Report

Key Insights (All Cities | 2023–2025)

1. **San Francisco** generated the **highest total revenue**, followed by **New York** and **Los Angeles**. All cities reached peak revenue in **May 2023**, with a gradual decline observed toward **April 2025**.
2. In terms of booking volume, **New York** recorded the **highest number of bookings** among the three cities.
3. **New York** achieved the **highest average review score (4.54)**, followed closely by Los Angeles (4.53) and San Francisco (4.48).
4. Analysis indicates that **cancelled bookings have little to no significant impact on guest review scores**.
5. Paid or settled transactions account for only **29%** of total potential revenue, indicating that the majority of bookings remain unpaid or pending.
6. The high proportion of unsettled payments suggests a potential issue in revenue collection, highlighting the need for improved payment monitoring and follow-up processes.